Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–IV (NEW) EXAMINATION – WINTER 2021

Subject Code:3140603 **Date:01/01/2022**

Subject Name:Structural Analysis-I

Time:10:30 AM TO 01:00 PM **Total Marks: 70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			MARKS
Q.1	**(a)**	Compute Static and kinematic indeterminacy of the structures shown in figure : 1.	**04**

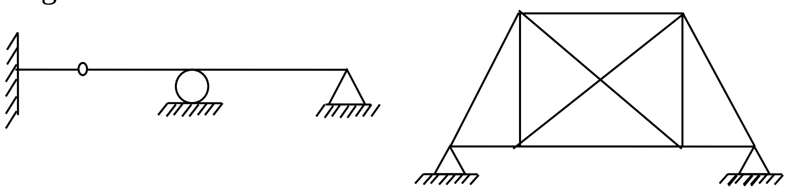

Figure : 1

	(b)	Distinguish between truss and frame.	**03**
	(c)	Analyze the beam shown in figure: 2 by moment distribution method and draw bending moment diagram.	**07**

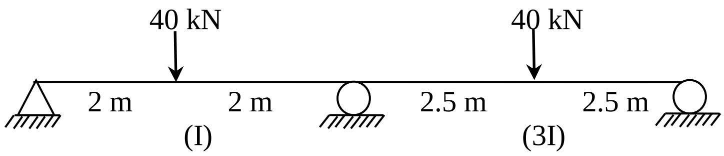

Figure: 2

Q.2	**(a)**	Differentiate between long and short column.	**03**
	(b)	Derive the expressions for Hoop and longitudinal stresses in a thin cylinder.	**04**
	(c)	For a three hinged parabolic arch having rise of 5 m, span of 25 m and loaded by a point load of 100 kN at 10 m from left support, calculate the maximum bending moment.	**07**

OR

	(c)	Analyze the portal frame shown in figure: 3 and draw shear force and bending moment diagram.	**07**

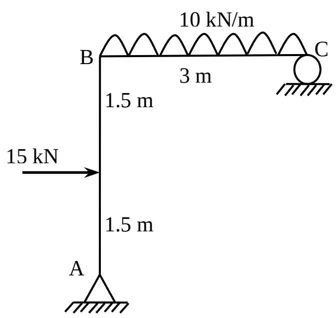

Figure: 3

Q.3	**(a)**	Derive relationship between strain energy due to gradual load and strain energy due to sudden load.	**03**
	(b)	Describe the Relationship between slope, deflection and radius of	**04**

curvature.

(c) Determine central deflection and slope at supports for a simply supported beam of length ‘l’ carrying a point load of ‘w’ at centre of the span. Take EI = constant. (Use Macaulay’s method) **07**

OR

Q.3 (a) Derive an expression for strain energy due to bending. **03**

(b) Explain gradual load, sudden load and load applied with impact. **04**

(c) A cantilever beam of span ’l’ is subjected to udl ‘w’ on the entire span. Find the slope and deflection at the free end by Conjugate beam method. **07**

Q.4 (a) Define conjugate beam. What are the different kinds of a support condition in conjugate beam? **03**

(b) Define slenderness ratio. With the help of sketches, explain effective length of column for different end conditions. **04**

(c) A rectangular column section ABCD (Figure: 4) having side AC=BD=450 mm and AB=CD=300 mm carries a compressive load of 230 kN at corner C. determine stress at each corner A,B,C,D also draw stress distribution diagram for each side. **07**

Figure: 4

OR

Q.4 (a) Derive equation for maximum and minimum stresses in rectangular section. **03**

(b) Write the assumptions and limitations of Euler’s formula. **04**

(c) Define column and strut. Show that for no tension in the base of a short column, the line of action of the load should be within the middle third. **07**

Q.5 (a) Write fixed end moment for a fixed beam of length ‘l’ carrying following loading on it **03**

(i) Central point load ‘w’

(ii) Uniformly distributed load ‘w’ over entire span

(iii) Point load ‘w’ at a distance ‘a’ from left support and ‘b’ from right support.

(b) Draw shear force and bending moment diagram for a fixed beam of length ‘l’ carrying a point load ‘w’ at centre. **04**

(c) Analyze the fixed beam shown in figure: 5 and draw bending moment diagram. **07**

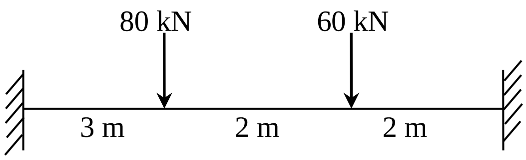

Figure : 5

OR

Q.5 (a) Differentiate between direct stress and bending stress with example. **03**

(b) Derive the equation of fixed end moment in a fixed beam of span ‘L’ having one of the support rotate clockwise θ. **04**

(c) A fixed beam AB of span L carried a UDL of w per meter length over entire span. Support B settles by ‘δ’ during application of load. Calculate the settlement ‘δ’, so that there is no fixed end moment at B. **07**

****